Creating Everyone Everywhere Affordable Broadband Access

Updated Framework & Case Study: Whatcom County

by Dan and Jamie Douglass

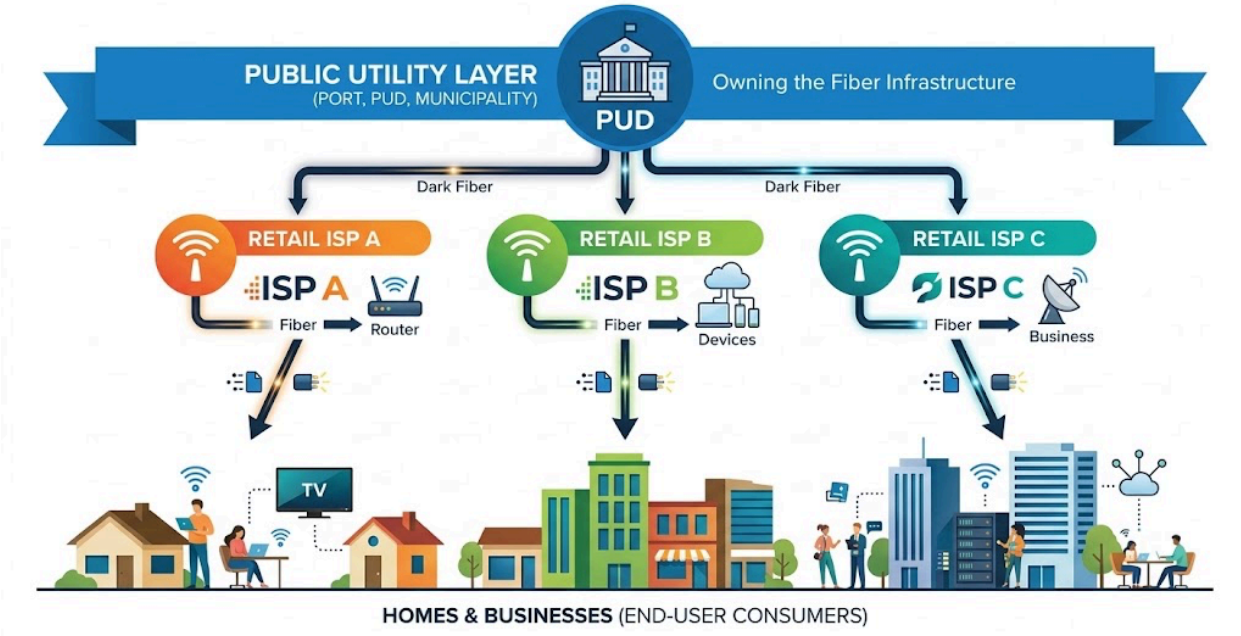
(Updated 2026 / Original July 24, 2022)

Executive Summary

This paper updates our original framework for creating affordable fiber-optic broadband access by integrating real-world lessons from regional deployments, specifically in Whatcom County, Washington. The benchmark for essential service remains **100 Mbps symmetrical** (download/upload) at or below **$70/month**, scaling to **1 Gbps symmetrical** at **$100/month**.

To achieve universal coverage, physical network infrastructure must be treated as a publicly owned, regulated utility using an **open-access model**. Decomposing networks into neighborhood-scale projects—such as Local Utility Districts (LUDs) or public-private partnerships—allows a structured "divide and conquer" strategy. Implementation success relies on multi-source funding, rigorous total-cost-of-ownership (TCO) analyses (e.g., aerial vs. buried fiber), and dynamic open-access retail competition among private Internet Service Providers (ISPs).

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1. Updated Internet Service & Affordability Standards

Definition: Affordability

No more than **$70/month for 100 Mbps symmetrical** service and **$100/month for 1 Gbps symmetrical** service. Public Utility Districts like Kitsap PUD have demonstrated that open-access networks can deliver service at these price targets. Like electric and water utilities, low-income households should be supported through income-qualified subsidies, including free basic tiers for qualifying families. These rate benchmarks are grounded in surveys of publicly owned fiber-optic networks across Washington State.

Definition: Internet Service Standards

The Federal Communications Commission (FCC) historically defined high-speed broadband as 25 Mbps download and 3 Mbps upload—a standard that categorized many inadequately served rural households as "fully served." While Washington State Senate Bill 5715 raised statewide targets to 100 Mbps down and 20 Mbps up, asymmetric standards remain fundamentally flawed. They assume end-users primarily consume short bursts of incoming data.

Modern applications—including hybrid remote work, high-definition tele-health, cloud data synchronization, interactive distance learning, and smart-home

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infrastructure—require substantial upload capacity. Symmetrical transmission treats user devices as peer-to-peer participants rather than passive consumers.

- **Baseline Standard:** 100 Mbps symmetrical (download/upload).
- **Scaling Target:** 1 Gbps symmetrical with less than 30 ms latency to support long-term regional economic viability.

2. Fiber Optic Infrastructure & Network Architecture

Recommendation: Physical link-layer infrastructure must use fiber optics whenever possible.

Fiber optics remains the only true future-proof technology for broadband deployment. Constrained only by the speed of light, physical glass fiber can transport virtually unlimited data volumes. Upgrading network capacity on a fiber network is accomplished by swapping optoelectronic terminal equipment in software and central offices without replacing physical cables or disturbing underground rights-of-way. Non-fiber alternatives (such as fixed wireless or satellite) should serve strictly as interim bridges while physical fiber backbones and last-mile connections are constructed.

Recommendation: Network infrastructure should be publicly owned and regulated.

Treating the physical fiber link-layer like public roads, power lines, or water mains resolves market failures inherent to private telecommunication monopolies. Private profit expectations cause commercial ISPs to favor densely populated urban centers while neglecting low-density rural corridors. Under a public open-access framework:

- **The Public Entity** (Port, PUD, or Municipality) owns, builds, and maintains dark fiber, conduit, and physical rights-of-way.
- **Private Retail ISPs** lease capacity on the open-access network, competing dynamically for subscribers based on customer service, speed tiers, and price.

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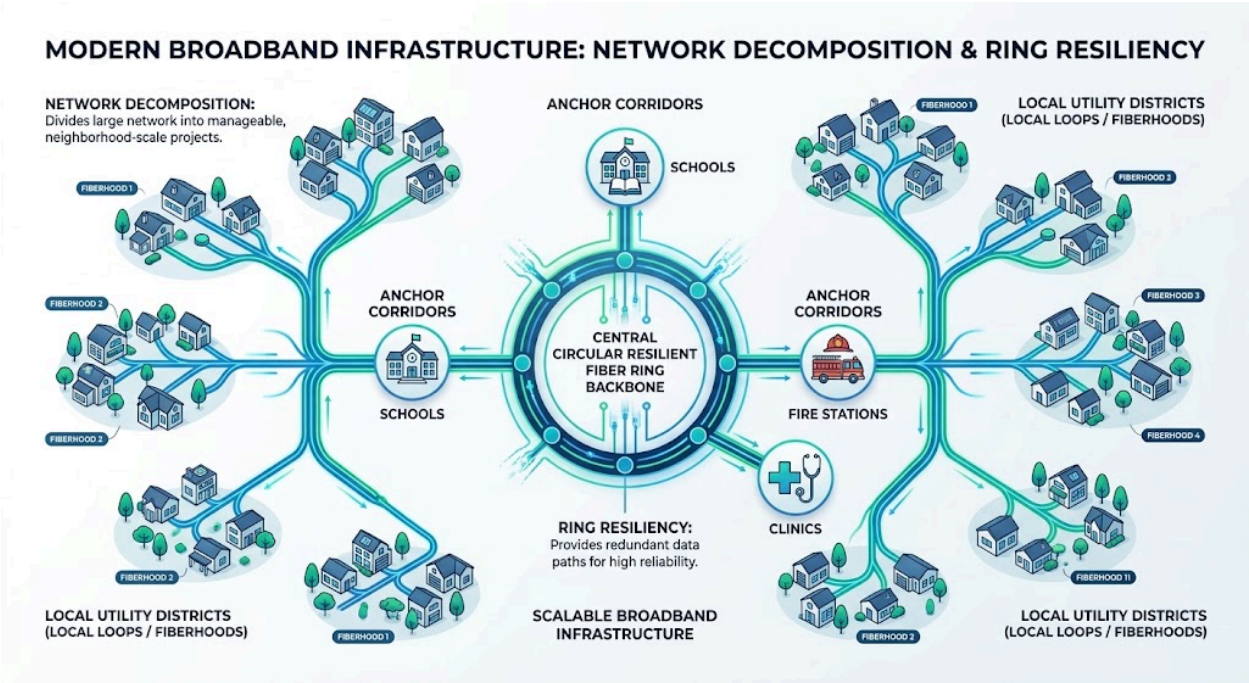
Recommendation: Decompose networks into local loops using a "divide and conquer" neighborhood structure.

Large regional builds are best executed by breaking geographic areas into independent, manageable local loop projects:

1. **Neighborhood Local Utility Districts (LUDs) / Fiberhoods:** Property owners opt-in to fund last-mile builds through long-term tax assessments or localized queue structures.

2. **Anchor Corridors:** Extending backbone fiber along primary routes to municipal buildings, fire stations, and schools creates immediate drop points for nearby residential branches.

3. **Ring Resiliency:** Moving beyond linear "point-to-point" connections to full ring architecture ensures continuous uptime during severe weather or physical line breaks.



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Recommendation: Leverage multi-source project financing.

Project execution requires combining multiple funding channels:

- **Property Owner Assessments:** Long-term, low-interest 20-year assessments attached to property taxes.
- **Public Grants & Subsidies:** Combining state and federal broadband grants (e.g., WSBO, CERB, ARPA, BEAD) with local match funds.
- **Anchor Tenant Commitments:** Using guaranteed municipal usage fees to underwrite capital infrastructure loans.

3. Case Study: Whatcom County Fiber Initiatives

Whatcom County, Washington, provides a clear case study of open-access public models operating alongside commercial market responses.

Pandemic Drivers & Demand Shift

The COVID-19 emergency transformed broadband in Whatcom County from an economic convenience into a critical public utility. Homework gap crises in rural school districts (such as Mt. Baker and Ferndale) required temporary parking-lot Wi-Fi hotspots. Post-pandemic, demand shifted permanently from temporary access to high-reliability, symmetrical connections driven by remote work, digital health services, and municipal management.

Public Sector Accomplishments & Challenges

The partnership between the **Port of Bellingham** and **Whatcom PUD #1** demonstrates both the strengths and hurdles of public infrastructure builds:

- **Successes:** Leveraged state and federal grants (WSBO, CERB, ARPA) alongside county matches to build open-access networks in unserved areas

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including East Nooksack, North Lynden, and North Ferndale.

- **Failures & Bottlenecks:** Experienced major cost overruns and timeline delays in geographically complex pockets like **Point Roberts**. Permitting backlogs, utility pole "make-ready" replacement fees, and supply chain inflation forced significant reductions in planned household connections.

Municipal & Commercial Landscape

- **City of Bellingham:** Adopted a long-term **Fiber Network Comprehensive Plan** focused on municipal operational needs, ring resiliency, public safety connectivity, and structured 10-year capital expansion.

- **Commercial Expansion:** Private ISPs (such as Ziply Fiber) accelerated capital investments across lower-density municipalities (Lynden, Ferndale, Everson, Maple Falls), creating parallel competition alongside public open-access footprinting.

4. Tools and Implementation Considerations

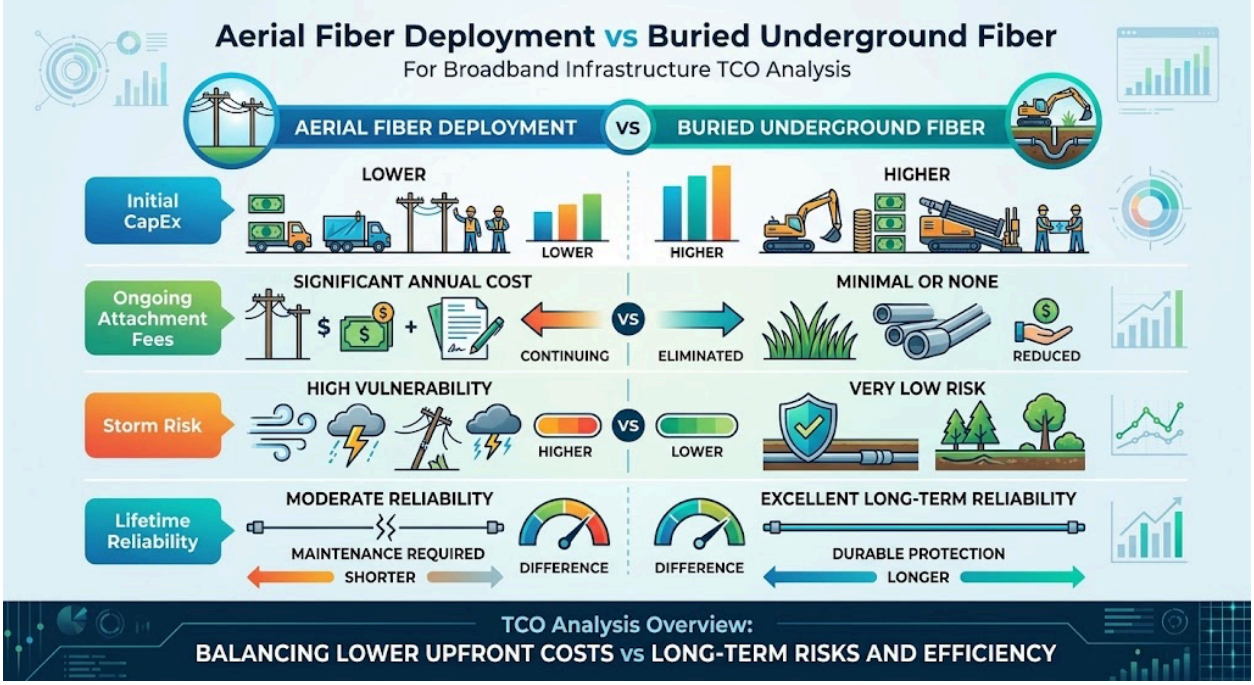
Recommendation: Evaluate Total Cost of Ownership (TCO) for Aerial vs. Buried Fiber.

Decisions between buried and aerial fiber construction must be driven by lifetime cost-benefit and reliability analyses rather than initial capital outlay alone:

- **Aerial Installation:** Offers lower initial capital outlay by utilizing existing utility poles, but carries high ongoing pole lease ("make-ready") fees and high vulnerability to windstorms, falling branches, and physical disruptions.

- **Buried Fiber:** Requires higher initial capital for directional boring or soft-plow trenching, but delivers significantly lower long-term risk, minimal operational maintenance, and higher overall uptime.

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Recommendation: Implement independent network performance testing and mapping.

Crowdsourced, independent speed-testing platforms are required to continuously monitor network health and map actual service delivery. Testing tools must evaluate sustained, normal multi-protocol user operations rather than short-burst prioritizing algorithms used by commercial ISPs to inflate advertised speeds.

Recommendation: Enforce strict Service Level Agreements (SLAs) for Retail ISPs.

Public wholesalers operating open-access networks must require participating retail ISPs to sign SLAs mandating:

- Full adherence to **Net Neutrality** principles.

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- Symmetrical download/upload speeds with no data cap restrictions or overage penalties.

- Month-to-month billing options with clear separation between residential and commercial pricing tiers.

Conclusion

Combining public infrastructure ownership with private retail competition establishes an equitable framework for universal broadband access. As demonstrated in Whatcom County, public funding initiatives address rural market failures, while private ISPs deliver competitive service options. By pairing open-access network architecture with strict service standards, multi-source financing, and neighborhood-level deployment strategies, communities can guarantee affordable, future-proof connectivity for all residents.

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